Beyond Department Boundaries: Centrality and Communication Communities in an Email Network

Course: DSA5101 Introduction to Big Data | **Evaluation:** Anonymous Submission | **NUS Matriculation:** A0352547Y

Project Title	**Beyond Department Boundaries: Centrality and Communication Communities in an Email Network**
Dataset & Source	Email-Eu-core network, Stanford Network Analysis Platform (SNAP) (1,005 members, 42 departments)
Algorithm Classes	1. PageRank Centrality (Lecture 6) 2. Graph Clustering / Louvain Community Detection (Lecture 7)
Research Question	To what extent do emergent communication communities align with formal department boundaries, and what structural roles do highly central individuals play in connecting disparate communities?

1 Research question and dataset

Formal organizational charts reflect institutional hierarchies, administrative responsibilities, and official reporting channels. However, day-to-day operational exchange, collaboration, and information dissemination frequently transcend designated organizational units. Investigating whether emergent communication clusters mirror administrative boundaries—and how centrally positioned individuals bridge or reinforce those clusters—provides critical insights into organizational functioning, cross-departmental silos, and coordination bottlenecks.

This project investigates the **Email-Eu-core** network published by the Stanford Network Analysis Platform (SNAP) [1]. Generated from email logs of a large European research institution over an 18-month observation window, the dataset captures interactions among researchers, staff, and leadership. Crucially, each individual is labeled with one of 42 ground-truth administrative departments. The network contains 1,005 institution members and 25,571 directed edges (u, v), representing at least one email sent from individual u to individual v. Edge weights representing email frequencies are not provided; edges indicate binary reachability.

We formalize three coherent research questions: **RQ1 (Prominence vs. Popularity):** How does directed PageRank centrality differ from simple in-degree popularity, and what topological mechanisms govern ranking disagreements? **RQ2 (Community-Department Alignment):** Do unsupervised communication communities discovered via Louvain modularity optimization follow the 42 administrative department boundaries? **RQ3 (Role of Central Individuals & Degree Confounding):** Do structurally prominent individuals act as cross-community boundary spanners or insular internal hubs, and does PageRank provide structural information beyond total node degree?

2 Data preparation

Data Integrity and Universe of Nodes: Raw edge and department label files were acquired from SNAP and verified against cryptographic hashes (SHA-256: 4b47acdb... for edges, e5abe5b4... for labels). Parsing revealed exactly 25,571 edge rows and 1,005 label rows, with zero malformed entries and zero duplicate directed edges. Department identifiers span $0 \le \text{dept} \le 41$, and node identifiers span $0 \le \text{node} \le 1004$.

Self-Loop Removal and Dual Graph Representation: An edge audit revealed 642 self-loops (u, u), where an individual emailed themselves (draft storage, personal reminders, or automated calendar echoes). Retaining self-loops distorts network analytics: in PageRank, a self-loop traps teleported random walk mass; in community detection, self-loops inflate internal edge density without representing interpersonal exchange. To ensure reproducibility and methodological rigor, we maintain dual representations: **(1) G_{raw}:** The raw directed graph containing all 1,005 nodes, 25,571 directed edges, and 642 self-loops, matching published SNAP statistics exactly. **(2) G_{analysis}:** The cleaned interpersonal directed graph containing all 1,005 nodes and exactly 24,929 interpersonal directed

edges ($25,571 - 642$). Crucially, the 19 singleton individuals whose only raw edges were self-loops are strictly preserved as degree-0 nodes with their ground-truth department labels to prevent selection bias.

Undirected Projection ($G_{\text{undirected}}$): Louvain modularity optimization assumes symmetric adjacency. We construct an unweighted undirected projection $G_{\text{undirected}}$ from G_{analysis}, collapsing reciprocal directed links (u, v) and (v, u) into a single undirected edge $\{u, v\}$. This projection preserves all 1,005 nodes (including the 19 isolates) and contains exactly 16,064 undirected edges. Reciprocal links account for 8,865 pairs ($17,730$ directed edges, 71.12%, matching the 0.7112 directed reciprocity of G_{analysis}), while 7,199 links are unreciprocated (28.88%). Directional initiation asymmetry is collapsed into binary reachability.

Network Representation	Nodes (|V|)	Edges (|E|)	Self-Loops	Density Formula & Value	WCC Giant	SCC Giant	Reciprocity (r)
G_raw (Published SNAP)	1,005	25,571	642	ρ = |E|/|V|² = 0.025317	986 (20 WCCs)	803 (203 SCCs)	0.6934 (69.3%)
G_analysis (Cleaned Directed)	1,005	24,929	0	ρ = |E|/[|V|(|V|-1)] = 0.024706	986 (20 WCCs)	803 (203 SCCs)	0.7112 (71.1%)
G_undirected (Louvain Projection)	1,005	16,064	0	ρ = 2|E|/[|V|(|V|-1)] = 0.031841	986 (20 CCs)	— (Undirected)	1.0000 (Symmetric)

Table 1: *Topological properties and structural invariants across graph representations.*

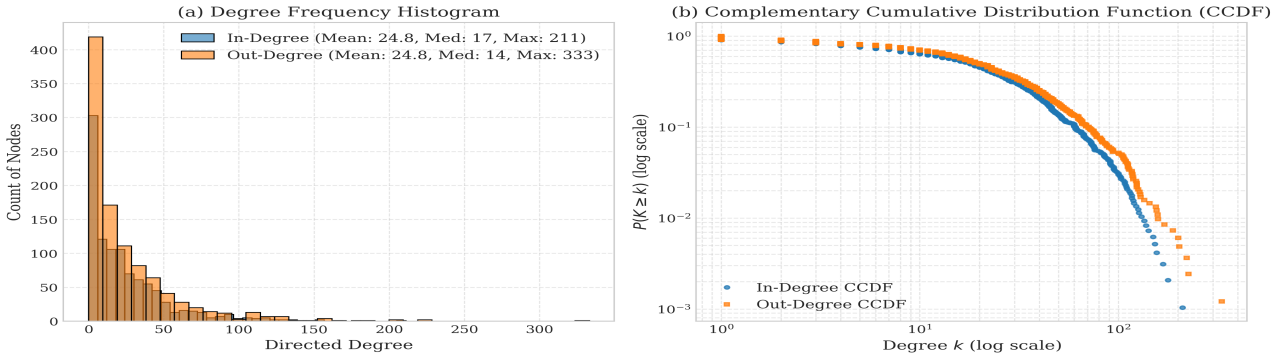


Figure 1: *Exploratory degree distributions on G_{analysis}. (a) Linear in-degree and out-degree histograms (mean = 24.8, median = 17.0 and 14.0). (b) Log-log complementary cumulative distribution function (CCDF) showing heavy-tailed communication activity.*

3 Methods and how they connect

Algorithm Class 1: PageRank Centrality (Lecture 6): Directed email networks represent endorsement flows where an incoming link (u, v) indicates that sender u acknowledges receiver v. PageRank models a random surfer who follows outgoing links with probability d (damping factor) and teleports uniformly across all $|V|=N$ nodes with probability $1-d$: $$\mathbf{r} = d \mathbf{P}^T \mathbf{r} + \frac{1 - d}{N} \mathbf{1}$$ where $\mathbf{P}_{uv} = A_{uv} / d_{\text{out}}(u)$ if $d_{\text{out}}(u) > 0$. When a node has zero out-degree (a dangling node, such as the 19 preserved isolates and 15 pure recipients), standard NetworkX dangling mass redistribution redistributes its entire random walk probability uniformly across all N nodes at each iteration. This guarantees convergence without absorbing stationary probability. We set primary parameters: damping factor $d = 0.85$, uniform personalization, and convergence tolerance $\text{tol} = 10^{-6}$. Deterministic tie-breaking is enforced by sorting by descending PageRank and ascending node ID.

Algorithm Class 2: Louvain Modularity Optimization (Lecture 7): To detect emergent communication communities without supervision, we apply the Louvain algorithm to $G_{\text{undirected}}$. Louvain greedily optimizes Newman-Girvan modularity: $$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \gamma \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$ where $m = 16,064$ edges, k_i is node i's undirected degree, c_i is its community, and γ is the resolution parameter (baseline $\gamma = 1.0$, random seed 42). Louvain operates in two alternating phases: local modularity-gain node moves, followed by community aggregation into meta-graphs, repeating until modularity plateaus.

Community Quality Metrics: We assess structural cohesion using modularity Q and conductance $\phi(S)$. For a community S, conductance measures boundary cut relative to volume: $$\phi(S) = \frac{|\partial(S)|}{\min(\text{vol}(S), \text{vol}(V \setminus S))}$$ where $\partial(S) = \{\{u, v\} \in E \mid u \in S, v \notin S\}$ and $\text{vol}(S) = \sum_{u \in S} k_u = 2|E(S, S)| + |\partial(S)|$. Conductance evaluates boundary cut size relative to total degree volume (not the percentage of links, since internal edges contribute twice to volume). Lower conductance indicates a sharper, more self-contained boundary.

External Department Alignment: We benchmark discovered communities against formal departments using Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI). Baseline department modularity Q_{dept} and departmental conductance evaluate how well administrative boundaries delineate actual communication.

How the Methods Connect (Integrated Boundary Analysis): Rather than treating centrality and clustering as isolated exercises, we bridge them directly to answer RQ3. For every node v, we project its PageRank score against its boundary bridging behavior in $G_{\text{undirected}}$: $$\text{CrossCommunityCount}(v) = |\{u \in N(v) \mid c_u \neq c_v\}|, \quad \text{CrossCommunityFraction}(v) = \frac{\text{CrossCommunityCount}(v)}{k_v}$$ For isolated singletons ($k_v = 0$), the fraction is formally undefined and recorded as NaN. By evaluating both absolute count and fraction, we distinguish true global boundary spanners (high PageRank, high count, high fraction) from insular internal hubs (high PageRank, high count, low fraction) and peripheral bridges (low PageRank, low count, high fraction).

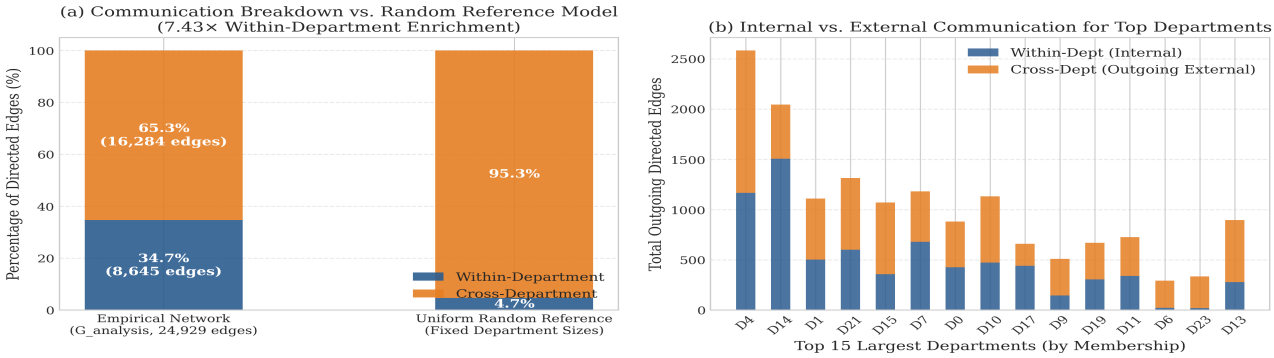


Figure 2: *Inter- vs. intra-departmental communication. Observed within-department edges (34.68%) exceed the uniform random ordered-pair null model ($p_{\text{null}} = 4.67\%$) by $7.43\times$, confirming strong departmental homophily amidst dominant (65.32%) cross-department communication.*

4 Evaluation and results

RQ1 Results: PageRank vs. In-Degree Centrality: Across all 1,005 nodes, PageRank and in-degree correlate strongly (Spearman $\rho_s = 0.9828, p < 10^{-300}$). Top-$k$ set overlap is high: 90.0% for $k=10$ (Jaccard = 0.818), 85.0% for $k=20$ (0.739), and 92.0% for $k=50$ (0.852). Prominence in Email-Eu-core is heavily degree-aligned. However, PageRank reveals critical structural disagreements driven by endorsement quality:

• *PageRank-Boosted (Node 451, Dept 14):* Possesses an in-degree of only 8 (rank 659, min rank 647), yet achieves PageRank 0.003117 (rank 37), a shift of $\Delta R = +622$. Senders endorsing Node 451 include Node 365 ($PR = 0.00306$, rank 39, out-degree 1), transferring 100% of its endorsement mass directly to Node 451. PageRank correctly values concentrated endorsements from prominent senders.

• *In-Degree-Boosted (Node 83, Dept 17):* Has in-degree 85 (rank 45) but falls to PageRank rank 56. Its senders have a high mean out-degree of 79.1, heavily diluting their endorsement mass ($12.8\times$ lower contribution per link).

• *Self-Loop Robustness:* In G_{raw}, self-loops artificially elevated the 19 singletons from rank 987 to rank 380 through reflexive trapping ($ ho_s = 0.8732$ between G_{raw} and G_{analysis}). Cleaning self-loops is mandatory.

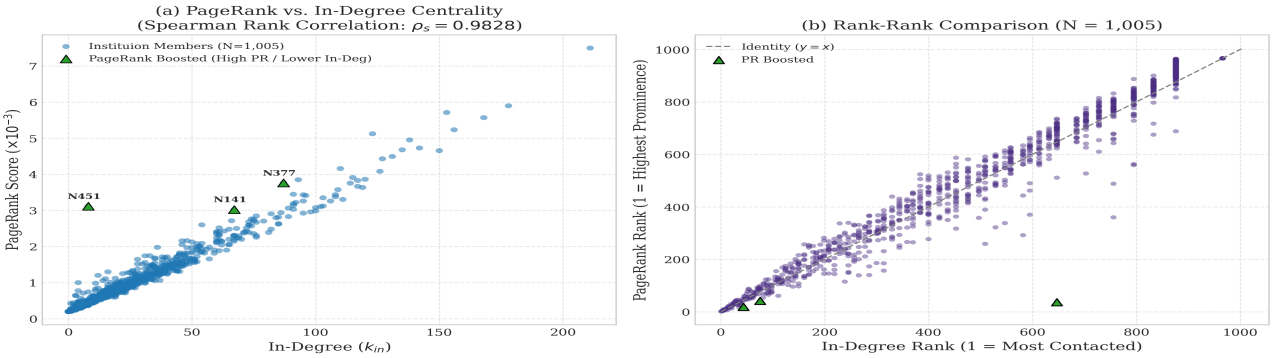


Figure 3: *PageRank vs. In-degree centrality on G_{analysis}. (a) Global scatter plot ($\rho_s = 0.9828$) highlighting Node 160 (Rank 1), Node 451 (+622 PageRank boost), and Node 83 (in-degree diluted). (b) Rank shifts ΔR for the Top 25 individuals.*

RQ2 Results: Louvain Community Detection vs. Administrative Departments: Louvain modularity optimization ($\gamma = 1.0, \text{seed} = 42$) discovers 28 communities: 9 non-trivial cohesive communities spanning 986 nodes, plus 19 isolated singletons (sizes: C0: 190, C1: 157, C2: 145, C3: 127, C4: 125, C5: 109, C6: 91, C7: 32, C8: 10). Louvain achieves modularity $Q_{\text{Louvain}} = 0.4130$ and median conductance 0.4201 (mean 0.4475, min 0.2780 in C6). By contrast, partitioning the same graph by official departments yields substantially lower modularity $Q_{\text{dept}} = 0.2880$ and severe boundary leakage (median departmental conductance 0.8644, mean 0.7871). External concordance is moderate: Normalized Mutual Information $NMI = 0.5946$, and Adjusted Rand Index $ARI = 0.3505$. Organic communication does not adhere to official administrative lines; instead, communication consolidates smaller departments into multi-departmental alliances while fragmenting specific units across clusters.

Partition Scheme	Clusters	Modularity (Q)	Median Conductance (ϕ)	Mean Conductance (ϕ)	NMI vs. Depts	ARI vs. Depts
Louvain Baseline (γ=1.0, seed=42)	9 (+19 isolates)	0.4130	0.4201 (C6: 0.2780)	0.4475	0.5946	0.3505
Official Departments (SNAP)	42 departments	0.2880	0.8644 (Dept 3: 0.301)	0.7871	1.0000 (Self)	1.0000 (Self)
Louvain (γ=0.8, Coarser)	6 (+19 isolates)	0.4105 (Q_obj: 0.459)	0.3812	0.4015	0.5272	0.2223
Louvain (γ=1.2, Finer)	10 (+19 isolates)	0.4051 (Q_obj: 0.383)	0.4410	0.4628	0.6247	0.4321

Table 2: *Quality metrics and external alignment across community detection configurations and official departments.*

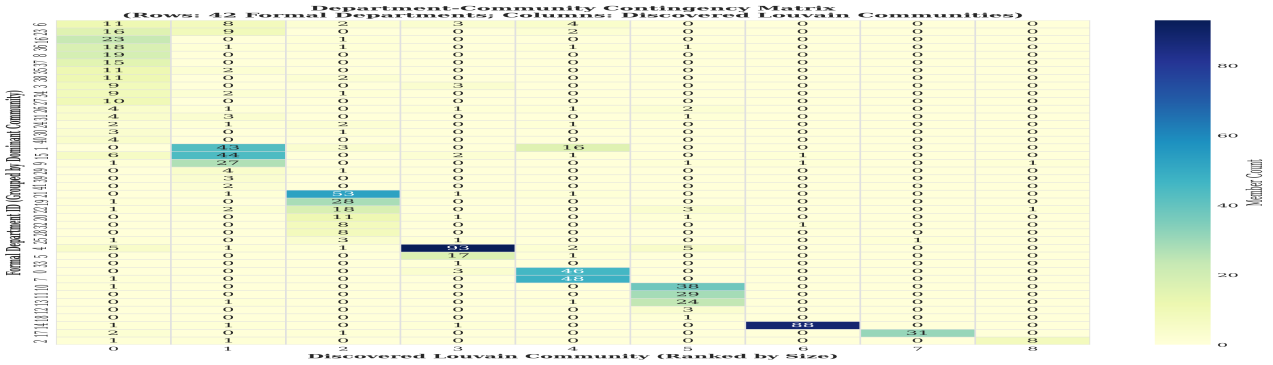


Figure 4: *Department-community contingency matrix heatmap for the 15 largest departments. Organic communication groups consolidate multiple departments (e.g. Community 0 unites Depts 4, 14, 36) while fragmenting others (e.g. Dept 14 splits across C0, C2, C6).*

RQ3 Results: Central Individuals as Boundary Spanners: Partitioning nodes into the Top 10% PageRank tier ($N=101$) and Remaining 90% tier ($N=904$):

• *Cross-Community Connections Count:* Top 10% maintain a mean of **67.88** cross-community edges (median 59.0) vs. **7.94** (median 4.0) for remaining nodes—an **8.55× higher volume** (Mann-Whitney $U = 88,970.0, p = 6.88 \times 10^{-56}$).

• *Cross-Community Connection Fraction:* Top 10% dedicate a mean of **55.79%** of their links outside their primary community (median 58.10%) vs. **26.56%** (median 25.00%) for remaining non-isolated nodes ($N=885$), more than double the proportion (Mann-Whitney $U = 77,037.5, p = 4.59 \times 10^{-33}$).

Degree Confounding and Stratified Analysis: Total undirected degree is a major confounder, correlating strongly with PageRank ($\rho_s = 0.9614$) and cross-community count ($\rho_s = 0.9155$). Controlling globally for degree yields a slightly negative partial rank correlation ($\text{Partial } \rho_s = -0.1803, p = 1.15 \times 10^{-8}$) because receiving concentrated endorsements inside dense core clusters boosts PageRank without boundary spanning. However, degree stratification resolves this dynamic: exactly **100 of the 101 Top 10% PageRank nodes (99.0%) reside in the highest degree quartile Q4** ($d \ge 45, N=247$). Within Q4, Top 10% PageRank nodes exhibit a $2.72\times$ higher cross-community count (**68.56 vs. 25.17**, $p = 2.10 \times 10^{-27}$) and significantly higher cross-boundary fraction (**56.34% vs. 41.30%**, $p = 5.69 \times 10^{-12}$). PageRank reliably distinguishes broad multi-cluster brokers from internal hubs among well-connected individuals.

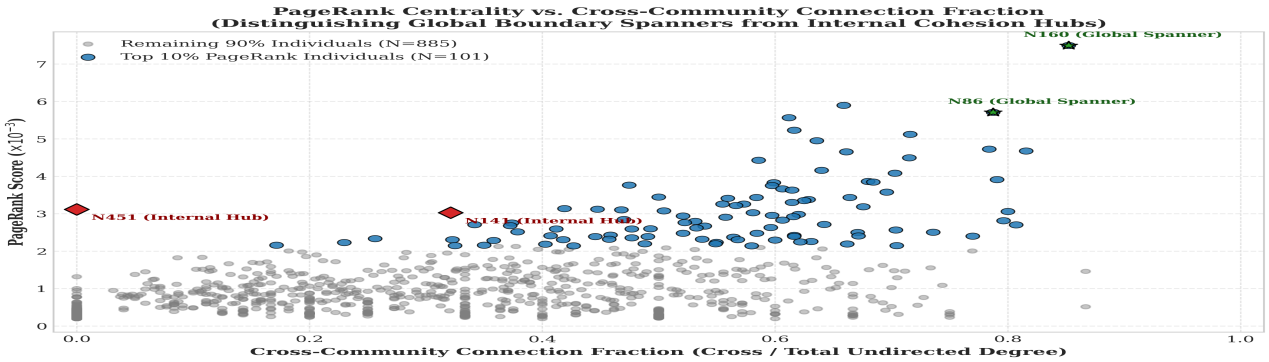


Figure 5: *Integrated analysis of PageRank vs. cross-community fraction with degree colormap and annotated structural archetypes: Node 160 (Global Boundary Spanner), Node 451 (Internal Hub), and Node 644 (Peripheral Bridge).*

Structural Archetype	Focal Node	Dept / Comm	PageRank (Score / Rank)	In-Degree (Rank)	Undirected Degree	Internal / Cross Edges	Cross Fraction	Comms Spanned
Global Boundary Spanner	Node 160	Dept 36 / Comm 4	0.007495 (Rank 1)	211 (Rank 1)	345	51 internal / 294 cross	**85.22%**	8 external comms
Internal Community Hub	Node 451	Dept 14 / Comm 6	0.003117 (Rank 37)	8 (Rank 659)	13	13 internal / 0 cross	**0.00%**	0 external comms
Peripheral Bridge	Node 644	Dept 25 / Comm 7	0.000514 (Rank 633)	11 (Rank 606)	15	2 internal / 13 cross	**86.67%**	5 external comms

Table 3: *Descriptive structural archetypes illustrating distinct topological configurations in Email-Eu-core.*

5 Insights and limitations

Substantive Organizational Insights: Our findings demonstrate that informal communication reorganizes official academic hierarchies. While departmental homophily is strongly enriched ($7.43\times$ above random expectation), nearly two-thirds (65.32%) of directed communications cross departmental boundaries. Unsupervised community detection reveals that departments consolidate into multi-departmental research clusters with sharp communication boundaries (conductance 0.420 vs. 0.864 for departments). Furthermore, elite prominence is overwhelmingly associated with boundary spanning: individuals in the top 10% PageRank tier maintain an $8.55\times$ higher cross-community degree and dedicate more than double their ties across cluster borders. However, this bridging role is not uniform: individuals such as Node 451 achieve high centrality through insular prestige without boundary spanning.

Methodological Limitations:

1. *Network Autocorrelation & Inferential Tests:* Network observations violate the independent and identically distributed (i.i.d.) assumption due to dyadic reciprocity (71.12%) and transitive closure. Reported Mann-Whitney U test p-values are descriptive indicators of distributional separation rather than

formal causal proofs.
2. *Binary vs. Weighted Edges:* The dataset lacks email frequencies, durations, and content. A single administrative announcement is topologically indistinguishable from frequent daily exchanges.
3. *Non-Causal Interpretations:* Network position reflects structural embedding, not formal organizational authority, leadership effectiveness, or deliberate brokerage intent. Case studies represent structural archetypes, not verified organizational roles.

6 Reproducibility information

Software and Hardware Environment: Analysis executed under Python 3.10.6 on an AMD Ryzen 7 processor with 16 GB RAM (fully operable on ≈ 8 GB RAM laptops). Dependencies: NetworkX 3.4.2, SciPy 1.15.3, scikit-learn 1.5.2, pandas 2.3.3, NumPy 1.26.4, matplotlib 3.10.9, seaborn 0.13.2. Pinned versions are documented in requirements.txt.

Random Seeds and Determinism: Random seed 42 is fixed across all stochastic components (Louvain optimization, NumPy random state). Tie-breaking in rankings follows a strict deterministic protocol (sorting by score descending, then node ID ascending).

Standalone Notebook: The accompanying analysis.ipynb notebook is completely self-contained, importing no local modules. It automatically downloads raw data from SNAP, verifies SHA-256 checksums, executes all analyses, and regenerates all reported tables and figures from a fresh kernel.

References

Note: Per official DSA5101 instructions, the reference list is excluded from the eight-page limit. Appendices are not included.

[1] H. Yin, A. R. Benson, J. Leskovec, and D. F. Gleich, 'Local higher-order graph clustering,' in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 555–564. [SNAP Email-Eu-core dataset].

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